

DGA Status Report

Dissolved gas analysis — status assessment, IEEE C57.104-2019 Figure 2
G057/PE/1/04/MT02_X/01

Status 3 - Probably suspicious

Report generated	2026-09-07 13:16	Latest sample	2025-12-17	Samples used	79	History from	2015-12-16
O2/N2 section	<=0.2	Age band	10-30 (26 yr)	Table 4 period	10-24 months	Multi-point rate	available

1. Verdict

Status 3 on CH4, C2H6, C2H4 above the Table 2 (95th percentile) level and CH4, C2H6 rising faster than Table 4 allows.

Required action — Identify the fault (Duval triangle and pentagon, Rogers ratios), place the unit under increased surveillance, add supporting tests and consult a transformer specialist.

Age taken from the year of manufacture 2000 (cmms-map).

2. What drove the status

- CH4 is 158 ppm, above the Table 2 (95th percentile) level of 150 ppm.
- CH4 is rising at +23.2 ppm/year over 3 samples, against a Table 4 allowance of 10 ppm/year.
- C2H6 is 216 ppm, above the Table 2 (95th percentile) level of 175 ppm.
- C2H6 is rising at +21.6 ppm/year over 3 samples, against a Table 4 allowance of 9 ppm/year.
- C2H4 is 100 ppm, above the Table 2 (95th percentile) level of 95 ppm.

3. Decision path (Figure 2)

Step	Question	Answer	Detail
Levels vs Table 1	Is every measured gas below its 90th percentile level?	No	CH4, C2H6, C2H4 above
Change vs Table 3	Is every change since the previous sample within the allowance?	Yes	all within
Rate vs Table 4	Is every multi-point rate within the allowance?	No	CH4, C2H6 over
Levels vs Table 2	Is any measured gas above its 95th percentile level?	Yes	CH4, C2H6, C2H4 above
Result	Figure 2 outcome	Status 3	Probably suspicious

4. Gas-by-gas evidence

Gas	Latest ppm	T1	T2	Level	Change since previous	T3	Verdict	Rate ppm/yr	T4	Rate
H2	25.0	100	200	within	+1.0	40	within	-6.8	20	within
CH4	158.0	90	150	above T2	-53.0	30	within	+23.2	10	over
C2H6	216.0	90	175	above T2	+25.0	25	within	+21.6	9	over
C2H4	100.0	50	95	above T2	+3.0	20	within	-6.4	7	within
C2H2	0.0	1	4 ‡	within	-7.0	any incr.	within	—	any incr.	n/a
CO	211.0	900	1100	within	+10.0	250	within	+21.2	100	within
CO2	1815.0	10000	14000	within	+70.0	2500	within	-22.4	1000	within

Rates are the slope of a least-squares fit over the 3 samples from 2024-05-07 to 2025-12-17 (19 months).

5. Limits applied

Column selected by the standard: O2/N2 <=0.2, age 10-30, Table 4 period 10-24 months.

Table	H2	CH4	C2H6	C2H4	C2H2	CO	CO2
Table 1 — 90th percentile level	100	90	90	50	1	900	10000
Table 2 — 95th percentile level	200	150	175	95	4	1100	14000
Table 3 — change between samples	40	30	25	20	any incr.	250	2500

Table	H2	CH4	C2H6	C2H4	C2H2	CO	CO2
Table 4 — rate (ppm/yr)	20	10	9	7	any incr.	100	1000

6. Sample history used

Date	H2	CH4	C2H6	C2H4	C2H2	CO	CO2	O2	N2
2015-12-16	93	589	282	615	0	247	3308	—	—
2015-12-30	85	557	300	645	0	225	3345	—	—
2016-01-20	89	556	271	583	0	257	3068	—	—
2016-08-05	141	837	364	815	0	378	3941	—	—
2016-10-03	130	781	344	781	0	363	3748	—	—
2016-10-20	32	225	118	258	0	77	1181	—	—
2016-11-01	37	261	157	346	0	85	1740	—	—
2016-11-16	39	277	148	329	0	101	1757	—	—
2016-11-24	31	264	154	348	0	80	1846	—	—
2016-12-08	51	265	141	308	0	128	1658	—	—
2017-02-08	50	330	165	379	0	120	2022	—	—
2017-03-31	48	400	178	415	0	169	2260	—	—
2017-06-07	74	471	202	488	0	196	2537	—	—
2017-08-25	72	401	185	472	0	129	2176	—	—
2017-11-17	85	433	201	515	0	143	2262	—	—
2018-03-06	106	519	246	644	0	163	2555	—	—
2018-04-13	137	656	260	714	0	226	2824	—	—
2018-07-13	140	718	272	751	1	226	2779	—	—
2018-07-24	122	699	271	737	0	218	2756	—	—
2018-08-01	120	681	279	755	0	216	2812	—	—
2018-08-07	126	679	283	761	0	217	2816	—	—
2018-08-14	125	699	294	793	0	227	2899	—	—
2018-08-21	130	695	274	747	1	226	2745	—	—
2018-08-28	122	653	267	730	0	216	2684	—	—
2018-09-11	127	652	277	755	0	219	2762	—	—
2018-10-03	131	713	262	725	0	218	2661	—	—
2018-10-17	125	700	256	707	0	206	2587	—	—
2018-11-10	144	715	254	718	0	246	2645	—	—
2018-12-01	123	693	284	773	0	232	2857	—	—
2018-12-20	129	688	253	698	0	230	2636	—	—
2019-01-16	113	699	294	788	0	229	2938	—	—
2019-02-06	80	509	228	622	0	124	2332	—	—
2019-02-08	108	671	279	752	0	218	2844	—	—
2019-03-06	86	549	255	672	0	163	2556	—	—
2019-03-28	99	719	283	767	0	213	2909	—	—
2019-04-08	101	681	266	720	0	223	2756	—	—
2019-05-02	95	701	253	690	0	217	2614	—	—
2019-07-10	89	665	256	688	—	214	2611	—	—
2019-08-05	92	686	277	733	—	225	2784	—	—
2019-09-02	83	663	257	683	—	229	2610	—	—
2019-10-14	70	647	253	666	—	212	2542	—	—
2019-12-06	—	—	5	9	—	2	90	—	—
2019-12-07	—	4	5	11	—	3	95	—	—
2019-12-07	—	—	6	13	—	3	117	—	—
2019-12-10	—	10	7	20	—	9	252	—	—

Date	H2	CH4	C2H6	C2H4	C2H2	CO	CO2	O2	N2
2019-12-13	—	12	8	22	—	12	340	—	—
2019-12-19	—	19	11	30	7	22	558	—	—
2019-12-27	—	19	9	30	—	21	543	—	—
2020-01-16	—	23	11	37	—	24	653	—	—
2020-02-07	—	39	17	54	—	51	963	—	—
2020-03-02	0	34	14	50	—	29	829	—	—
2020-04-29	0	35	17	49	—	37	787	—	—
2020-05-27	8	57	21	67	—	65	1134	—	—
2020-07-16	8	61	21	65	—	75	1234	—	—
2020-08-19	0	46	24	57	0	39	992	—	—
2020-09-29	10	65	29	70	—	72	1223	—	—
2020-11-17	9	66	27	73	—	68	1194	—	—
2020-12-12	13	50	34	71	—	72	1194	—	—
2021-01-21	—	71	38	74	—	79	1229	—	—
2021-03-01	11	70	42	80	—	80	1227	1031	21598
2021-04-04	12	81	49	89	—	92	1372	—	—
2021-05-04	10	72	49	85	—	83	1291	—	—
2021-07-07	17	81	59	93	—	98	1434	—	—
2021-08-02	13	78	57	87	—	99	1373	—	—
2021-09-21	16	88	61	87	—	109	1366	—	—
2021-10-27	17	84	63	87	—	112	1359	—	—
2021-11-24	14	92	67	89	—	110	1392	—	—
2022-01-06	22	90	72	91	—	117	1440	—	—
2022-03-07	14	100	80	93	—	123	1469	—	—
2022-05-26	14	104	92	98	—	129	1528	—	—
2022-07-28	15	101	90	88	—	127	1394	—	—
2022-09-20	13	107	100	93	—	134	1463	—	—
2022-12-13	17	102	110	95	—	140	1471	—	—
2023-02-06	16	113	116	98	—	140	1518	—	—
2023-06-06	20	123	129	99	—	157	1575	—	—
2023-11-06	25	121	140	98	—	165	1579	—	—
2024-05-07	35	135	178	109	—	177	1833	—	—
2025-06-16	24	211	191	97	—	201	1745	2078	79790
2025-12-17	25	158	216	100	—	211	1815	—	—